SCIENCE 1.4 **Name:**_____

WORKSHEET FIVE **REACTIONS OF ACIDS -Word equations**

1. **Key reactions to remember**. Complete the three general key reactions of acids.

ACID + BASE ⟶ _____

ACID + METAL ⟶ _____

ACID + CARBONATE ⟶ _____

2. **Acid + Metal**

(a) Sulfuric acid + magnesium ⟶ _____

(b) Nitric acid + aluminium ⟶ _____

(c) Hydrochloric acid + zinc ⟶ _____

(d) Ethanoic acid + magnesium ⟶ _____

(e) sulfuric acid + zinc ⟶ _____

3. Acid + Carbonate

(a) Hydrochloric acid + sodium carbonate ⟶ _____

_____

(b) sulfuric acid + calcium carbonate ⟶ _____

_____

(c) Nitric acid + potassium carbonate ⟶ _____

_____

(d) Ethanoic acid + zinc carbonate ⟶ _____

_____

(e) Nitric acid + sodium bicarbonate ⟶ _____

_____

4. **Acid + Base**

(a) Hydrochloric acid + sodium hydroxide → _____

(b) Sulfuric acid + sodium hydroxide → _____

(c) Nitric acid + sodium hydroxide → _____

(d) Sulfuric acid + copper oxide → _____

(e) Hydrochloric acid + copper oxide → _____

(f) Nitric acid + zinc oxide → _____

(g) Ethanoic acid + calcium oxide → _____

(h) Hydrochloric acid + lead oxide → _____

(i) Sulfuric acid + copper hydroxide → _____

(j) Ethanoic acid + iron(II) hydroxide → _____

5. **Naming salts**

Name the salt formed when hydrochloric acid reacts with

(a) sodium hydroxide _____

(b) copper oxide _____

(c) lithium carbonate _____

(d) aluminium oxide _____

Name the salt formed when sulfuric acid reacts with

(a) calcium oxide _____

(b) lead carbonate _____

(c) iron(III) hydroxide _____

(d) potassium hydroxide _____

6. **Identifying gases produced**

(a) Describe how you test for hydrogen gas and say what happens in a positive test.

_____

(b) Describe how you test for carbon dioxide gas and say what happens in a positive test.

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